

Algorithm 1 SGDF: Online Filter Estimate Gradient (element-wise).

Input: θ_0 : initial parameter, $f(\theta)$: stochastic objective function

Parameter: $\{\alpha_t\}_{t=1}^T$: step size, $\{\beta_1, \beta_2\}$: attenuation coefficients, $\{\varepsilon\}$: numerical stability, $\{\gamma\}$: power scaling.

Output: θ_T : resulting parameters.

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1: Initialize:  $m_0 \leftarrow 0, s_0 \leftarrow 0$ 
2: while  $t \leq T$  do
3:    $g_t \leftarrow \nabla f_t(\theta_{t-1})$ 
4:    $m_t \leftarrow \beta_1 m_{t-1} + (1 - \beta_1) g_t$ 
5:    $s_t \leftarrow \beta_2 s_{t-1} + (1 - \beta_2)(g_t - m_t)^2$ 
6:    $\hat{m}_t \leftarrow \frac{m_t}{1 - \beta_1^t}, \hat{s}_t \leftarrow \frac{(1 - \beta_1)(1 - \beta_1^{2t}) s_t}{(1 + \beta_1)(1 - \beta_2^t)}$ 
7:    $K_t \leftarrow \frac{\hat{s}_t}{\hat{s}_t + (g_t - \hat{m}_t)^2 + \varepsilon}$ 
8:    $\hat{g}_t \leftarrow \hat{m}_t + K_t^\gamma (g_t - \hat{m}_t)$ 
9:    $\theta_t \leftarrow \theta_{t-1} - \alpha_t \hat{g}_t$ 
10: end while
11: return  $\theta_T$ 
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